

MANET - Routing Protocols

Module 41

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Hello friends I am engineer s Kumar from school of electrical engineering and information technology Punjab Agriculture University Ludhiana I welcome you all to the video lecture series of cyber and information security in this lecture we will study about routing protocols used in mobile adog network so now let us start before going into the details of routing protocol it is necessary to know about the routing protocol what it performs actually and what is their importance in the field of communication generally routing protocol represents the relationship or formula that is being used by routers to find the suitable way through which data can be forwarded routing protocol is also helpful in exchange of information between routers routing protocol plays a

vital role to adjust the network according to variable conditions in other words the routing protocol is the implementation of routing algorithm in the form of software or Hardware routing protocol uses different matrices to find out the path which is used to transfer the packet across the network next types of routing protocols there are three types of routing Protocols are used in mobile adhoc network reactive routing protocols proactive routing protocols and hybrid routing protocols first reactive routing protocols reactive routing Protocols are called On Demand routing protocols so these routing Protocols are called when they are needed and the routers are built these routers can be acquired by sending root request through the network disadvantage of this algorithm is that

it offers High latency in searching a network next proactive routing protocols the routing information about all the nodes is built and maintained by the proactive protocols the proactive routing Protocols are independent of whether or not the route is needed control messages are transmitted with periodically intervals even if there is no data flow still control messages are transmitted because these control messages proactive routing Protocols are not bandwidth efficient there are many advantages and disadvantages of proactive rting protocols one of its Advantage is that the nodes can easily get routing information and it easily starts a session the disadvantages are too much data kept by the nodes for rout maintainance when there is a particular

link Failure Its reform is too slow next hybrid routing protocol this approach is a hybrid approach of combining both benefits of reactive and proactive routing protocols the routing initiates using the proactive method of pre planned routes extra demand is then handled by reactive flooding this methodology is only suitable for some situations where traffic demand and the number of nodes can be determined beforehand first protocol is aodv AD hoc on demand distant Vector this is reactive protocol seek to set up routes on demand if a node wants to initiate communication with the node to which it has no route the routing protocol will try to establish such a route aodv protocol is described in RFC

3561 the philosophy in aodv like all reactive protocols is that topology information is only transmitted by nodes on demand when a node wishes to transmit traffic to a host to which it has no route it will generate a root request which is rreq message that will be flooded in a limited way to other nodes this causes control traffic overhead to be dynamic and it will result in an initial delay when initiating such communication a route is considered found when the RQM reaches either the destination itself or an intermediate node with a valid route entry for the destination for as long as route exists between two endpoints if it remains passive when the route becomes invalid or lost aodv will

again issue a request aodv defines three types of control messages for route maintenance first rreq a root request message is transmitted by a node requiring a route to a node as an optimization aodv uses an expanding ring technique when flooding these messages every rreq carries a time to live value that states for how many hops this message should be forwarded this value is set to a predefined value at the first transmission and increased at retransmissions retransmissions occur if no replies are received rreq frame format First Source address then Source sequence then broadcast ID then destination address then destination sequence Last Hop count next RRP a root reply message is unicasted back to the originator of rreq if the receiver is either the node

using the requested address or it has a valid route to the requested address the reason one can unicast the message back is that every route forwarding a rreq caches a route back to the originator the format of RRP is Source address destination address destination sequence hop count and lifetime where lifetime indicates how long a path can be used that is $r \leq r$ nodes monitor the link status of next hops in active routes when a link breakage in an active route is detected a root error message is used to notify other nodes of the loss of the link in order to enable this report mechanism each node keeps a precursor list which contains the IP addresses for each of its neighbors that are likely to use it as a

next hop towards each destination example the diagram to the left shows a setup of four nodes on a wireless network the circles illustrate the range of communication for each node because of the limited range each node can only communicate with the nodes next to it node one wishes to send a message node three node one's neighbors are nodes 2 and 4 since node one cannot directly communicate with node 3 node one sends a root request message the

root request is H by node four and node two when node one name receive the root request message they have two choices if they know a route to the destination or if they are the destination they can send a root reply message back to node one otherwise they will rebroadcast the root request to their set of neighbors the message keeps getting rebroadcast

until its lifespan is up if node one does not receive a reply in a set amount of time it will rebroadcast plus the request accept this time the root request message will have a longer lifespan and a new ID number all of the nodes use a sequence number in the root request to ensure that they do not rebroadcast a root request in the example node two has a root to node three and replies to the root request by sending out a root reply node four on the other hand does not have a root to node three so it rebroadcast the root request in the example node one is the forwarding a root reply to node four it notices that the root in the root reply has a better sequence number than the root in its routing list node one then replace the root it currently has with

the root in the root reply next root error messages root error messages allow a to adjust root Roots When nodes move around whenever a node receives root error it looks at the routing table and removes all the roots that contain the bad nodes the diagram to the left illustrates the three circumstances under which a node would broadcast a root error to its neighbors in the first scenario the node receives a data packet that it is supposed to forward but it does not have a root to the destination the real problem is not that the noes does not have a root the problem is that some other nodes thinks that the correct route to the destination is through that node in the second scenario the node receives a root error that cause at least one of its roots to become invalidated if it happens the node would

then send out a root error with all the new nodes which are now unreachable in the third scenario the node detects that it cannot communicate with one of its neighbors when this happens it looks at the root table for root that uses the neighbors for a next hope and marks them as invalid then it sends out a root error with the neighbors and the invalid Roots next routing protocol is osr osr is a proactive routing protocol and is also called as table driven protocol because it permanently stores and updates its routing table olsr keeps track of routing tables in order to provide a route if needed osr can be implemented in any adog network due to its nature osr is called as proactive routing protocol this protocol Works in collaboration with other nodes in Manet

through the exchange of topology information this exchange of information is done periodically olsr uses multi-point lays to avoid the broadcast of unnecessary packets retransmissions in a network a node broadcast a message periodically to its neighboring nodes this is done to compute the multi-point relay set as well as the exchange of information about the neighborhoods from the information about the neighborhood this node calcul Cal Ates the minimum set of one hop lay point that is needed to reach the two hope neighbors and this set is called the multi-point relay set osr differs from the link State protocols in two factors based

on the dissemination of routing information first is by construction that is only the multi-hop lay nodes of a node a need to forward updates about link state that are issued

by node a secondly the size of the link State update of node a is reduced because it only consist of those neighbors that selected nodes a as their multi-point lay node thus we can conclude that olsr reduce the link State protocol it is used in a network where nodes are densely deployed the olsr calculates the shortest path in such networks to an arbitrary destination following are the functions of protocol first neighbor sensing in wireless adog networks each node has to certify its nature of link with the neighbor nodes because of radio transmission in osr specifications there are two types of Link symmetric and asymmetric each node sends hello interval to its one of neighbors Hello message to achieve neighbor sensing

which have no not be forwarded when node send the message it contains node neighbor list and their link status which allow them to reduce the hold to Hope neighbor and their status afterward MPR selection is made and the list is added into hello messages in final step using this MPR list a MP selector list is constructed which contains a neighbors list which have selected it as ampere now the messages received from their ampere selectors will be forwarded by the nodes second ampere flooding the aim of multi-point relas is possibly control traffic flooding in ampere flooding ampers are selected in such a way that when a flooding message is transmitted by the Ampere set it must reaches all two hope neighbors Amper set

of a node n which is also represented as the smaller subset of symmetric one of neighbors of n having symmetric links with two hope neighbors of Amper flooding mechanism makes way to the elimination of transmission duplication as well as reception duplication is minimized by it next topology diffusion the objective of topology diffusion is to create routing tables using periodic topology control messages that is TC messages TC messages are circulated by each node with the nonn empty amp selector set to all Network nodes broadcasting at least links between itself and the nodes in its ampers Lector set to achieve topology diffusion these TC messages contain sufficient information which enable nodes first to construct their topology table and then

to derive their routing table the roots are using short path algorithm such as digra shortest path algorithm after their calculation and providing the best possible hopes number however there is always a need to recalculate the routing tables to update the route information as the tables are based on information concerning links to neighbors and topology which can be changed at any instant next protocol is Tora temporary order routing algorithm Tora is a hybrid routing protocol it is effective in solving the existing limitations in Mobile AOG networks due to the high mobility of nodes congestion is one of the major problems in manets traditional shortest path algorithm adoptive shortest path algorithm and Link state routing can not work properly in Mobile Stations it is difficult to

update the routing tables of each Dynamic nodes in to each node broadcast a cury packet and the recipients broadcast an update packet it Sports the loop free multiple route facilities using flat a non hierarchical routing algorithm it also provides better scalability to discover a route it uses a dag dag means directed a cyc graph and also uses a set of totally ordered Heights at all times in this approach information may only in One Direction hence it is only unidirectional there is no chance to fail in infinite low it performs three basic operations root Creations root maintainance and root deletion to perform these three basic operations this protocol uses three control packets such as cury update and

clear first root creation two control packets such as cury and updates are used to create a new route within the network a cury packet consist of a destination ID which is used for identifying the destination node and upd packet holds destination ID and height of the node each node maintains a root required flag which is initially unset and also maintains the time of broadcast for the last update packet within a network when a node has no directed link and an unset rout requested flag it broadcast a cury packet to its neighbor nodes and set its root required flag when any node receives a cury packet then it checks the following condition first if there is no down stream link and route required flag is unset it rebroadcast the cury control packet and

sets the root required flag next if there is no Downstream link and root required flag is set it discards the cury packet if the receiving node has at least one Downstream link with null height it sets its height and broadcast a upd packet if the receiving node has at least one Downstream link with the nonnull height first compares the time of Last Broadcast of update packet with the time of Link over which cury packet was received became active next when the link becomes active in a node it broadcast a update packet and discards the query packet if the root required flag is set of any node it broadcast a query packet when a node receives a update package from its neighbors it updates the entities of array height and proceeds through the following steps if the root required

flag is set node sets its high height and update all the entities in its link State array and unsets the root required flag then it broadcast a update packets with New Height if the root required flag is not set it just update the entities of the link State array and applies the root maintainance techniques next a root maintainance in Mobile hog Network Tora maintains the root when any topological change has occurred it also ensur the reestablishment of roots between nodes within finite amount of time the root maintenance is needed for a node when its height is non null if any neighbor node has null value that neighbor node will not be considered for this operation there are five different cases

available in Tora to root maintenance case one generate new reference level case two propagate the highest neighbors reference level case three reflect back a higher sub level case four partition detection and case five is generate new reference level in case one due to link failure the node lost its last downstep link if the node has any upam neighbor node it updates

its reference value otherwise it sets its height to null if the node has no Downstream link it propagates the reference level to its Upstream Neighbors which is shown in case two due to link reversal if the downstream links fail then the node reflects back the reference height with setting the reflection bit according to the case three if the set reference value is equal to the reference height of neighbor nodes the node has detected

the partition and has started to raise the root setting the height value to null which is shown in case four in case five generation of new reference value occurs when the node has experienced a link failure between the time of propagation of a reference level and reflects Su level next operation is root deletion in case four of root maintenance node sets its height which is determined by the direction of edges to the destination and updates all the entries of its link State array and broadcast control packet when a node receives a control packet it follows the following procedures first if the reference level in the control packet is matched with the reference level of that receiving node the node sets its height and sets the height for each neighbor to null it

also update all the entries in the link State arrays and broadcast control packet second if the reference level of the control packet does not match with the reference level of the node it sets its height for each neighbor and it also updates corresponding link State array entries at the end the height of each node in the portion of the network which was partitioned is set to null and erase all the invalid roots so friends this was all about routing protocols in Manet hope the concepts explained in this lecture were understandable and helpful hope to see you in the next lecture till then goodbye thank you